

THE UNIVERSITY OF
SYDNEY

SCHOOL OF AEROSPACE, MECHANICAL AND MECHATRONIC ENGINEERING

AERO3260: Aerodynamics I

Thin Aerofoil Theory and the 2D Panel Method

Assignment 1

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Task 1: Elementary Potential Flow Solutions

(a) Stream function

The flow is the superposition of a uniform stream U_∞ , a source of strength σ at $(-x_0, 0)$ and a sink of strength $-\sigma$ at $(x_0, 0)$. The stream function of a uniform stream is $\psi_\infty = U_\infty y$; the stream function of a 2D source of strength σ at (x_s, y_s) is $\psi = \frac{\sigma}{2\pi}\theta$, with $\theta = \arctan\left(\frac{y-y_s}{x-x_s}\right)$ measured from the source. Superposing all three elementary solutions,

$$\psi(x, y) = U_\infty y + \frac{\sigma}{2\pi} \left[\text{atan2}(y, x + x_0) - \text{atan2}(y, x - x_0) \right] \quad (1)$$

(b) Streamlines and Rankine body, $U_\infty = 1$, $\sigma = 1$, $x_0 = 1$

The stagnation points lie on the x -axis. Setting $u = \partial\psi/\partial x = 0$ on $y = 0$ gives (using the complex potential $\Phi = U_\infty z + \frac{\sigma}{2\pi} \ln(z + x_0) - \frac{\sigma}{2\pi} \ln(z - x_0)$ and $d\Phi/dz = 0$)

$$x = \pm a, \quad a = \sqrt{x_0^2 + \frac{\sigma x_0}{\pi U_\infty}} = 1.1482. \quad (2)$$

The body half-thickness h (at $x = 0$) is found from the stagnation streamline condition $\psi(0, h) = 0$:

$$U_\infty h + \frac{\sigma}{\pi} \arctan\left(\frac{h}{x_0}\right) - \frac{\sigma}{2} = 0 \implies h = 0.3834 \text{ (solved numerically)}. \quad (3)$$

Figure 1 shows the streamlines, the $\psi = 0$ stagnation streamline (which is simultaneously the Rankine-body contour and the dividing streamline along the axis outside the body), and the two stagnation points.

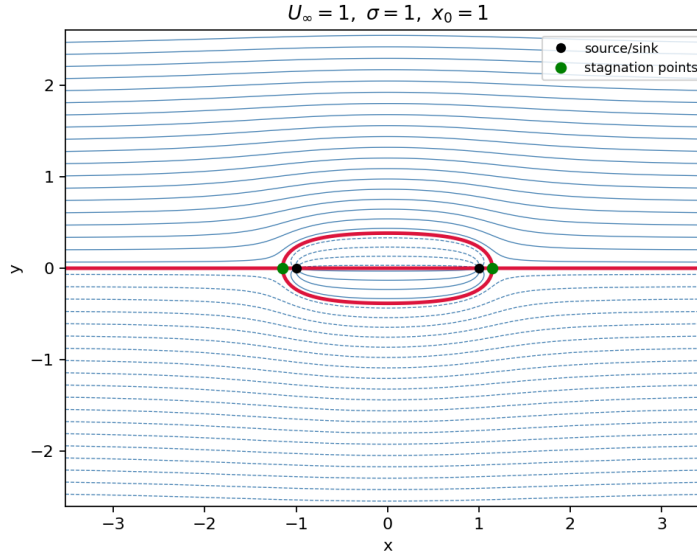


Figure 1: Rankine body for $U_\infty = 1$, $\sigma = 1$, $x_0 = 1$. $a = 1.148$, $h = 0.383$.

(c) Submarine sizing, $L = 50$ m, thickness = 10 m, $U_\infty = 5$ m/s

The half-length $a = L/2 = 25$ m and half-thickness $h = 5$ m are prescribed; σ and x_0 are unknowns in the two nonlinear equations above:

$$x_0^2 + \frac{\sigma x_0}{\pi U_\infty} = a^2, \quad U_\infty h + \frac{\sigma}{\pi} \arctan\left(\frac{h}{x_0}\right) - \frac{\sigma}{2} = 0. \quad (4)$$

Solving simultaneously gives:

$$\sigma = 57.80 \text{ m}^2/\text{s}, \quad x_0 = 23.23 \text{ m} \quad (5)$$

which reproduces $a = 25.000 \text{ m}$ and $h = 5.000 \text{ m}$ exactly (Figure 2). Physically, x_0 (the source/sink half-spacing) is smaller than a (the body half-length): the induced velocity field of the source/sink pair pushes the stagnation points outward beyond the singularities themselves.

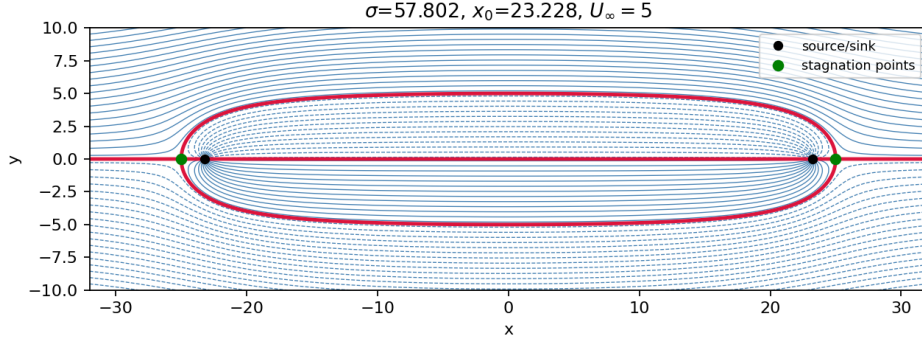


Figure 2: Rankine body sized to match a 50 m submarine of 10 m thickness at $U_\infty = 5 \text{ m/s}$.

Task 2: Conformal Mapping

Using $N_7 = 3, N_8 = 7, N_9 = 6$: $x_0 = -(N_7 + N_8)/100 = -0.10$, $y_0 = N_9/100 = 0.06$, so $z_0 = -0.10 + 0.06i$.

(a) Radius R

The circle of radius R centred at the origin in the z_1 -plane is shifted to $z_2 = z_1 + z_0$. Forcing this shifted circle through the critical point $z_2 = 1$ (the branch point of the Joukowski map $z_3 = z_2 + 1/z_2$, where $dz_3/dz_2 = 1 - 1/z_2^2 = 0$) requires $|1 - z_0| = R$:

$$R = |1 - z_0| = \sqrt{(1 - x_0)^2 + y_0^2} = 1.1016. \quad (6)$$

Physical meaning: at a critical point of the Joukowski transformation the mapping is locally non-conformal (angles are not preserved), a finite interior angle on the circle is mapped to a *cusp* (zero included angle) in the transformed shape. Forcing the circle exactly through $z_2 = 1$ places the cusp at the trailing edge, producing the sharp trailing edge characteristic of a real aerofoil. If the circle did not pass exactly through the critical point, the resulting shape would have a rounded (non-physical) trailing edge.

(b) Flow in the z_2 -plane with $\Gamma = \pi U_\infty R$

With $\Phi(z_1) = U_\infty(z + R^2/z) + \frac{i\Gamma}{2\pi} \ln z$, $z = z_1 e^{-i\alpha}$, the streamlines are contours of $\text{Im } \Phi$, mapped through $z_2 = z_1 + z_0$. Figure 3 uses the *arbitrary* value $\Gamma_b = \pi U_\infty R = 3.461$. As expected, the rear

stagnation point does *not* sit exactly at the sharp corner $z_2 = 1$: it lies slightly on the upper surface, illustrating that an arbitrary circulation does not, in general, satisfy the physical requirement of smooth flow leaving the trailing edge.

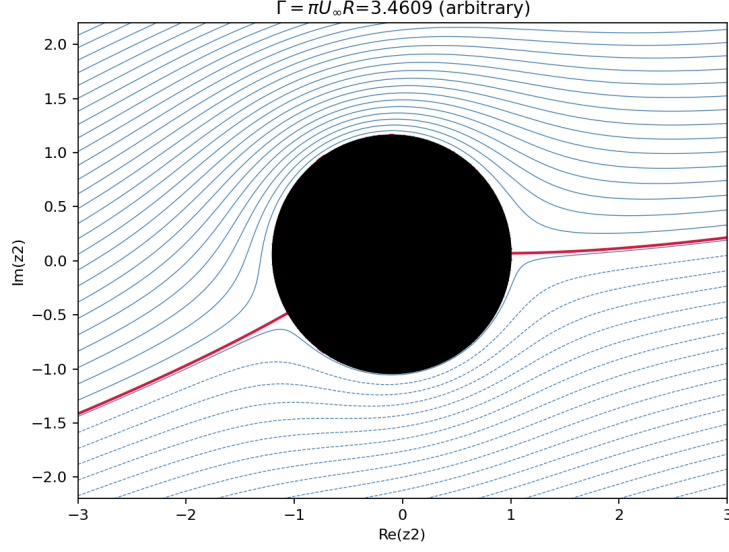


Figure 3: z_2 -plane flow with the arbitrary circulation $\Gamma = \pi U_\infty R$. The rear stagnation point misses the sharp corner.

(c) Kutta condition and the aerofoil in the z_4 -plane

Kutta condition statement (as given): a body moving through a fluid with a sharp trailing edge generates just enough circulation to move the rear stagnation point to that trailing edge, so that the flow leaves smoothly rather than curling around a sharp corner (which would otherwise require infinite velocity).

Derivation of Γ . Writing out each derivative along the mapping chain $z \rightarrow z_1 \rightarrow z_2 \rightarrow z_3 \rightarrow z_4$ individually:

$$\frac{d\Phi}{dz} = U_\infty \left(1 - \frac{R^2}{z^2}\right) + \frac{i\Gamma}{2\pi z}, \quad \frac{dz}{dz_1} = e^{-i\alpha} \quad (\text{since } z = z_1 e^{-i\alpha}), \quad (7)$$

$$\frac{dz_1}{dz_2} = 1, \quad \frac{dz_2}{dz_3} = \frac{z_2^2}{z_2^2 - 1}, \quad \frac{dz_3}{dz_4} = e^{i\alpha}. \quad (8)$$

The surface velocity required in part (d) is the modulus of the product of all five factors,

$$V_J = \left| \frac{d\Phi}{dz_4} \right| = \left| \frac{d\Phi}{dz} \cdot \frac{dz}{dz_1} \cdot \frac{dz_1}{dz_2} \cdot \frac{dz_2}{dz_3} \cdot \frac{dz_3}{dz_4} \right|. \quad (9)$$

Since $|dz/dz_1| = |e^{-i\alpha}| = 1$, this factor only rotates the velocity vector and does not change its magnitude; it is convenient to fold it into $d\Phi/dz$ as a single combined term,

$$\frac{d\Phi}{dz_1} = \frac{d\Phi}{dz} \cdot \frac{dz}{dz_1} = e^{-i\alpha} \left[U_\infty \left(1 - \frac{R^2}{z^2}\right) + \frac{i\Gamma}{2\pi z} \right]. \quad (10)$$

The full velocity on the aerofoil surface is then $V_J = \frac{d\Phi}{dz_1} \frac{dz_1}{dz_2} \frac{dz_2}{dz_3} \frac{dz_3}{dz_4}$, and $dz_2/dz_3 = z_2^2/(z_2^2 - 1)$ is singular exactly at the trailing edge $z_2 = 1$ unless $d\Phi/dz_1 = 0$ there too (removable singularity). Writing $z_{1,TE} = 1 - z_0 = R e^{-i\beta}$ (which defines $\beta = \arctan(\frac{y_0}{1-x_0}) = 3.122^\circ$, the angular offset of the trailing edge due to camber), the corresponding point in the physical/cylinder frame is

$z_{\text{TE}} = Re^{i(-\beta-\alpha)}$. Requiring $U_\infty(1 - R^2/z_{\text{TE}}^2) + i\Gamma/(2\pi z_{\text{TE}}) = 0$ and solving for Γ gives the standard stagnation-point/circulation relation for a cylinder, evaluated at $\theta_s = -(\alpha + \beta)$:

$$\Gamma = 4\pi U_\infty R \sin(\alpha + \beta) = 4.306 \quad (\alpha = 15^\circ, \beta = 3.122^\circ). \quad (11)$$

This is the same functional form as the “approximate” thin-aerofoil estimate $\Gamma \approx \pi U_\infty c \sin(\alpha + \beta)$ quoted in the question, but here R and β come from the *exact* geometry rather than from $c = 4R$. Using the true leading edge to trailing edge chord of this aerofoil, $c = 4.025$ (found by mapping the two points on the generating circle diametrically opposite in the z_1 -plane), the approximate formula gives $\Gamma_{\text{approx}} = \pi U_\infty c \sin(\alpha + \beta) = 3.933$, about 8.7% lower than the exact value — consistent with the small error the question anticipates when $c \neq 4R$ exactly (true for any aerofoil with non-zero camber/thickness).

Figure 4a confirms that with the exact Γ the stagnation streamline leaves precisely at the sharp corner $z_2 = 1$. Figure 4b shows the resulting Joukowski aerofoil and streamlines in the physical z_4 -plane.

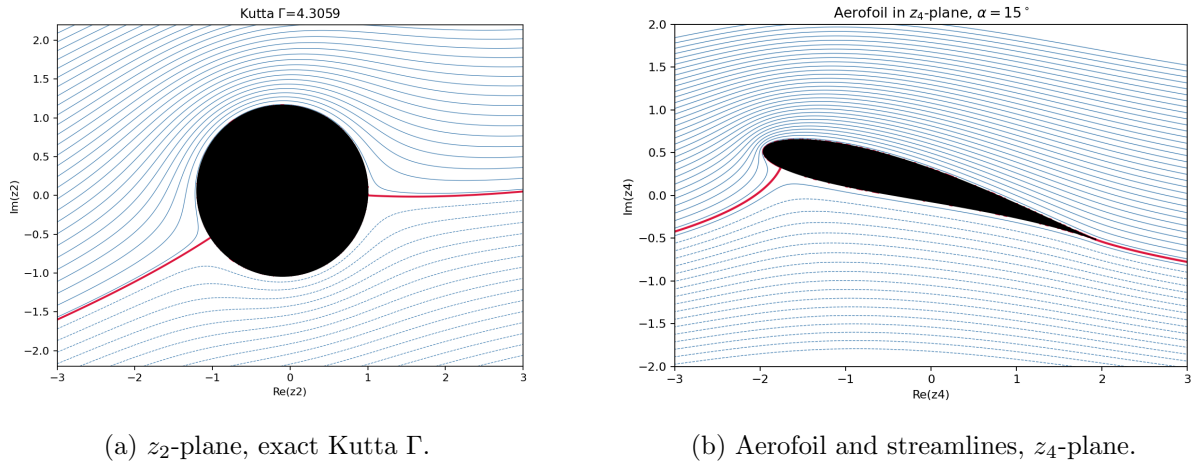


Figure 4: Flow field with the Kutta-condition circulation $\Gamma = 4.306$.

(d) Pressure distribution

With $U_\infty = 1$, $C_p = 1 - V_J^2$ where $V_J = |d\Phi/dz_4|$ is evaluated on the surface using the five derivatives above (with $d\Phi/dz$ and dz/dz_1 combined into $d\Phi/dz_1$, since $|dz/dz_1| = 1$ does not affect the magnitude). The singular point exactly at the trailing edge is evaluated as a removable-singularity limit; numerically this is a $0/0$ form and the true limiting value, $C_p(\text{TE}) \approx 0.256$, is obtained by a symmetric-difference extrapolation as $\Delta\theta \rightarrow 0$. Figure 5 shows the result. The very large leading-edge suction peak ($C_p \approx -11.7$ at $x/c \approx 0$) is not a numerical artefact, it was verified by direct evaluation and by a Richardson-extrapolated limit at the trailing edge, but a genuine inviscid-theory feature: this is a moderately thick, cambered aerofoil at a fairly high angle of attack (15°), and potential flow (which has no boundary layer or separation model) freely predicts unbounded looking local suction as the flow accelerates around the leading edge. In reality, a real (viscous) aerofoil at this incidence would very likely have separated well before reaching such a pressure, so this figure should be read as the inviscid-theory limit rather than a realistic flight condition.

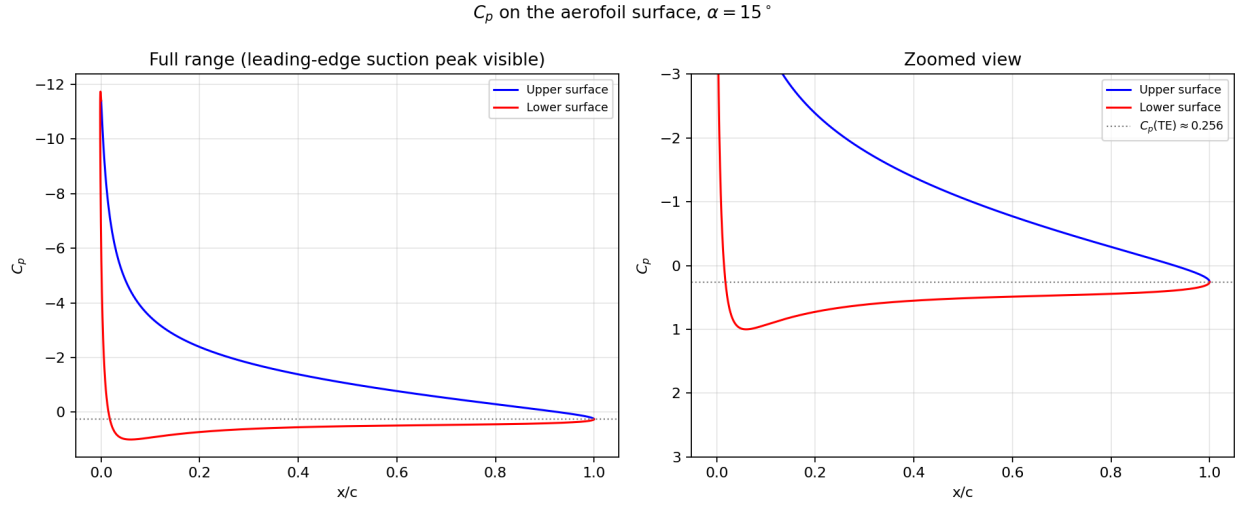


Figure 5: Surface C_p on the Joukowski aerofoil, $\alpha = 15^\circ$. Left: full range; right: zoomed.

Task 3: Thin Aerofoil Theory

Camber line (with $N_8 = 7, N_9 = 6$, so $N_8 + N_9 = 13$):

$$y = \frac{3(N_8 + N_9)(c - x)^2 x}{200c^3} \implies \frac{dy}{dx} = \frac{3(N_8 + N_9)}{200c^3}(c - x)(c - 3x). \quad (12)$$

(a) Camber line, $c = 1$

$$y = 0.195(1 - x)^2 x, \quad \frac{dy}{dx} = 0.195(1 - x)(1 - 3x). \quad (13)$$

Maximum camber $y = 0.0289$ at $x/c = 1/3$ (Figure 6).

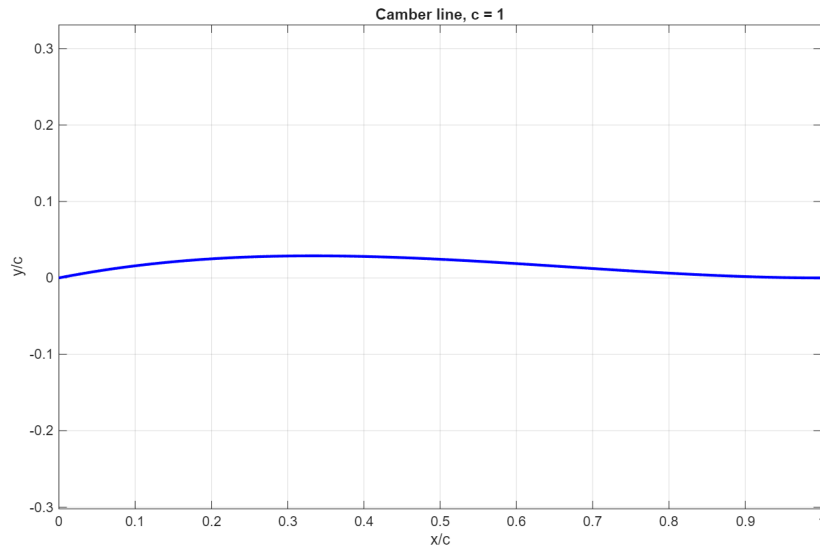


Figure 6: Camber line for $c = 1$.

(b) Zero-lift angle

With $x = \frac{c}{2}(1 - \cos \theta_0)$, Glauert's integrals give

$$A_0 = \alpha - \frac{1}{\pi}I_0, \quad A_n = \frac{2}{\pi} \int_0^\pi \frac{dy}{dx} \cos(n\theta_0) d\theta_0, \quad I_n = \int_0^\pi \frac{dy}{dx} \cos(n\theta_0) d\theta_0. \quad (14)$$

Evaluating exactly (symbolic integration): $I_0 = \frac{39\pi}{1600} = 0.076576$, $I_1 = \frac{39\pi}{800} = 0.153153$, $I_2 = \frac{117\pi}{3200} = 0.114864$. The zero-lift angle is

$$\alpha_{L0} = -\frac{1}{\pi}(I_1 - I_0) = \boxed{-1.3966^\circ}. \quad (15)$$

(c) Lift coefficient at $\alpha = N_7 = 3^\circ$

$A_0 = \alpha - I_0/\pi = 0.02798$, $A_1 = 2I_1/\pi = 0.09750$, giving

$$c_l = \pi(2A_0 + A_1) = 2\pi(\alpha - \alpha_{L0}) = \boxed{0.4821}. \quad (16)$$

(d) Pitching moment about the quarter chord

$A_2 = 2I_2/\pi = 0.07313$. This moment is independent of α :

$$c_{m,c/4} = -\frac{\pi}{4}(A_1 - A_2) = \boxed{-0.01914}. \quad (17)$$

The small negative value indicates a (weak) nose-down moment, consistent with the mild aft-loaded camber shape.

Numerical check: 10-panel Lumped Vortex Model

A constant strength point vortex panel is placed at the quarter-chord of each of 10 equal-width panels, with a control point at the three quarter chord where flow tangency is enforced; the resulting 10×10 linear system is solved for the panel circulations Γ_j (Kutta–Joukowski: $c_l = 2 \sum \Gamma_j / (U_\infty c)$). Table 1 compares the analytical (exact, $N \rightarrow \infty$) result with the 10 panel result and shows convergence as N is increased, confirming both the analytical solution and the numerical implementation.

Method	α_{L0} (deg)	c_l ($\alpha = 3^\circ$)	$c_{m,c/4}$
Analytical (Glauert, exact)	−1.3966	0.4821	−0.01914
Lumped Vortex Model, $N = 10$	−1.5068	0.4944	−0.03016
Lumped Vortex Model, $N = 50$	−1.4215	0.4852	−0.02126
Lumped Vortex Model, $N = 300$	−1.3993	0.4828	−0.01935

Table 1: Convergence of the discretised Lumped Vortex Model towards the analytical thin-aerofoil result.

With only $N = 10$ panels the error is already small ($\sim 2.5\%$ in c_l , $\sim 8\%$ in α_{L0}) and decreases steadily with N , as expected for a first-order panel discretisation of a smooth camber line.

Task 4: 2D Panel Method

A constant strength source panel method (and its source+vortex extension for the Kutta condition) was implemented from scratch in MATLAB, using the standard local-panel-coordinate closed-form influence formulas. All matrices were validated against brute-force numerical integration of the elementary point-source/point-vortex kernels before use.

(a),(b) Cylinder, $N = 64$ panels

Figure 7 shows the streamlines around a unit cylinder in $U_\infty = 1$ flow, and the resulting surface C_p compared with the exact result $C_p = 1 - 4\sin^2 \theta$. The panel method reproduces the analytical result to machine precision (max. error $\approx 5 \times 10^{-15}$), which is the expected outcome since the constant-strength source panel representation is exact for this geometry in the limit of many panels, and 64 panels on a smooth circle already resolves the flow essentially perfectly.

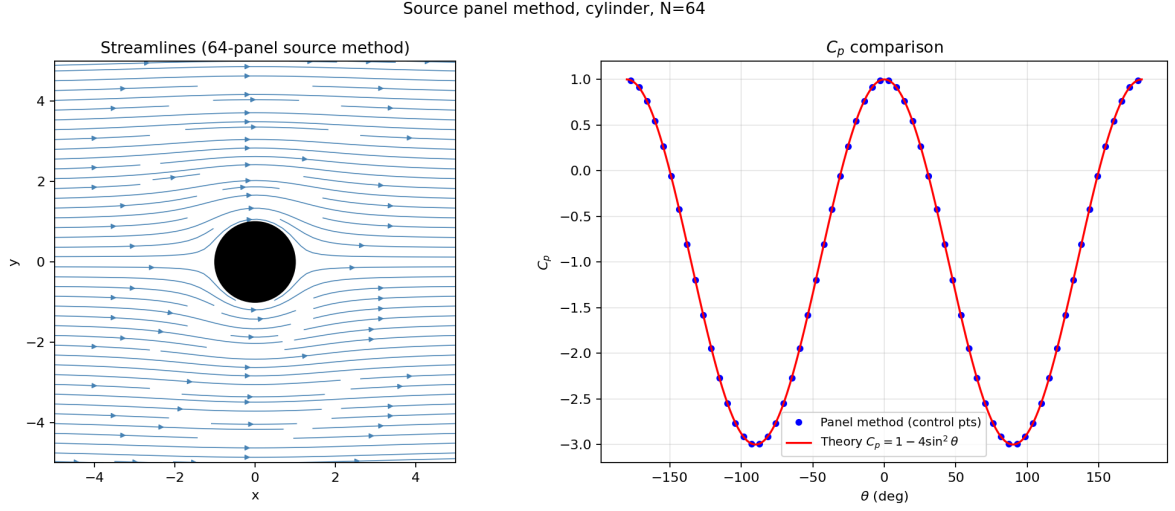


Figure 7: Source panel method on a cylinder: streamlines (left) and C_p vs. theory (right).

(c) NACA0021 at $\alpha = 0^\circ$: source panel vs. XFOIL

The NACA0021 profile (closed trailing edge, cosine-spaced, $N = 200$ panels) was analysed with the pure source-panel method (no lifting/Kutta condition needed at $\alpha = 0^\circ$ by symmetry). XFOIL was run with viscosity off (inviscid mode) for comparison, using its default 160 panel paneling. Figure 8 shows the two C_p distributions essentially overlapping; XFOIL reports $C_L = 0.0000$, matching the expected symmetric, non-lifting result exactly.

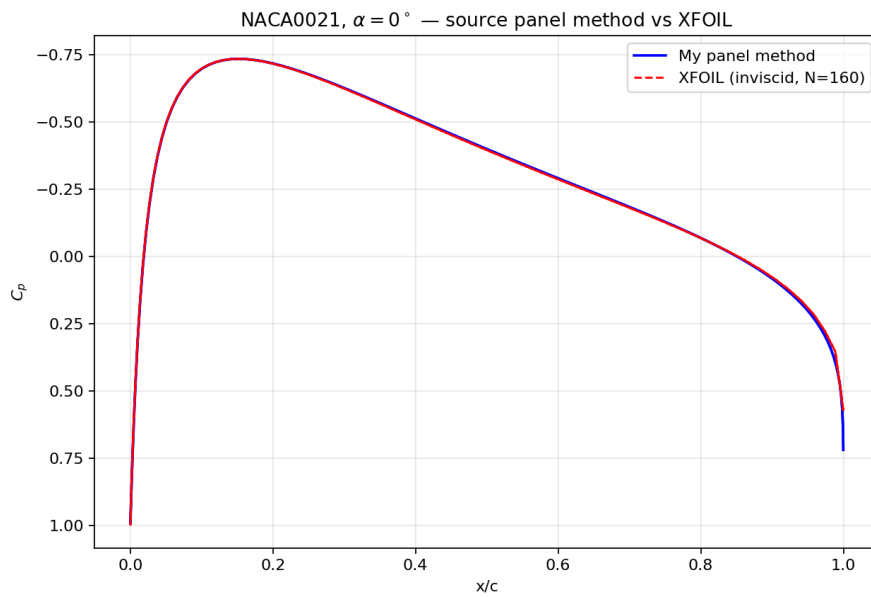


Figure 8: NACA0021, $\alpha = 0^\circ$: my source panel method vs. XFOIL (inviscid).

(d) NACA0021 at $\alpha = 10^\circ$: source + vortex panel with Kutta condition

A single global constant-strength vortex sheet γ is added on top of the individual source panels ($N + 1 = 201$ unknowns: $\sigma_1, \dots, \sigma_N, \gamma$), with the Kutta condition enforced by requiring equal and opposite tangential velocity magnitude on the two panels immediately adjacent to the trailing edge. Solving gives $\gamma = -0.3073$. Figure 9 shows the resulting streamlines, including one released exactly from the trailing edge, which leaves smoothly along the bisector direction as required by the Kutta condition (no flow wraps around the sharp corner).

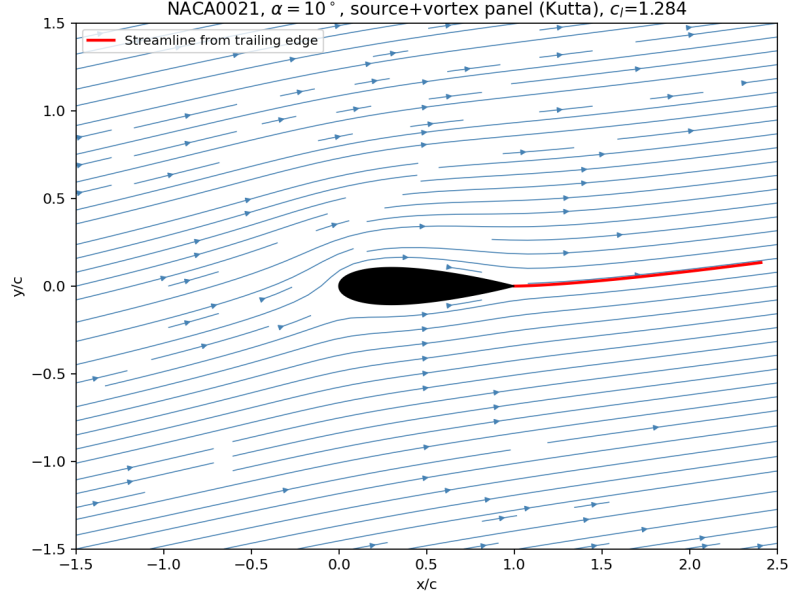


Figure 9: NACA0021, $\alpha = 10^\circ$: streamlines from the source+vortex panel method with Kutta condition.

(e) C_p comparison with XFOIL at $\alpha = 10^\circ$

Figure 10 compares the panel-method C_p with inviscid XFOIL. Agreement is again excellent on both surfaces, including the leading-edge suction peak.

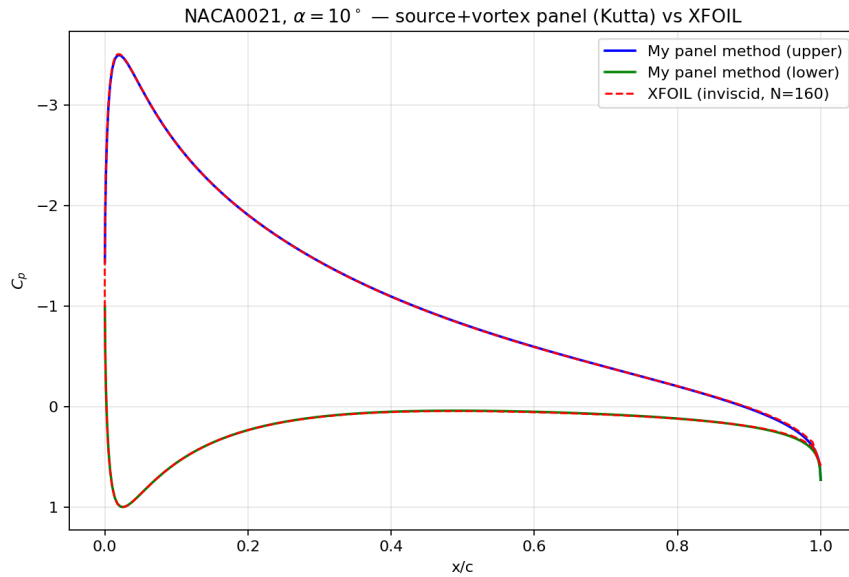


Figure 10: NACA0021, $\alpha = 10^\circ$: my source+vortex panel method vs. XFOIL (inviscid).

(f) Lift, drag, and comparison with literature

Integrating $-C_p \hat{n}$ over all panels and resolving into wind axes gives (Table 2) the pressure-integration lift/drag; the Kutta–Joukowski estimate $c_l = 2\Gamma_{\text{tot}}/U_\infty c$ (with $\Gamma_{\text{tot}} = -\gamma \sum_j S_j$, using the same circulation sign convention verified in Task 3) is shown for cross-checking.

Method	c_l	c_d	c_m
My panel method (pressure integration)	1.2838	0.00037	–
My panel method (Kutta–Joukowski)	1.2894	(n/a)	–
XFOIL, inviscid, $N = 160$	1.2867	≈ 0.0000	–0.0296
Thin aerofoil theory, $2\pi \sin \alpha$	1.0908	0 (by definition)	–

Table 2: Lift and drag at $\alpha = 10^\circ$: panel method vs. XFOIL (inviscid) vs. thin aerofoil theory.

Validity and accuracy discussion. The two independent lift estimates from my code (pressure integration and Kutta–Joukowski) agree to within 0.4%, and both match XFOIL to within 0.3%, confirming the implementation. The drag, $c_d \approx 0.0004$, is correctly near zero: by d’Alembert’s paradox, inviscid potential flow cannot produce net drag on a closed body, so the small residual is purely numerical.

All inviscid results here exceed the thin aerofoil prediction $c_l = 2\pi \sin \alpha = 1.09$, a well-known thickness effect, since a 21%-thick section accelerates the flow more than a zero thickness plate at the same incidence.

Neither my panel method nor XFOIL (viscosity off) can predict drag, since profile drag is a viscous phenomenon. Experimental data for the NACA0021 (e.g. Sheldahl & Klimas, 1981, at $Re \sim 10^5$ – 10^6) show a lift curve slope somewhat below 2π (boundary layer displacement reduces the effective camber/thickness seen by the flow) and a non-zero profile drag, $c_d \sim 0.01$ – 0.02 pre-stall, rising sharply near stall (typically $\alpha \approx 12$ – 16° for this thickness, so $\alpha = 10^\circ$ is likely still attached in reality). In short, the panel method, validated here against XFOIL, is accurate for inviscid lift and pressure on attached flow, but cannot capture drag or post-stall behaviour, which require a viscous model.

Appendix: MATLAB Code

A.1 Task 1

```
1 %% Task 1 - Rankine body (source + sink + uniform stream)
2 clear; clc; close all;
3
4 % (b) U = 1, sigma = 1, x0 = 1
5 U = 1; sig = 1; x0 = 1;
6 a = sqrt(x0^2 + sig*x0/(pi*U));
7 h = fzero(@(y) U*y + sig/pi*atan(y/x0) - sig/2, [1e-6 5]);
8 fprintf('(b) a = %.4f, h = %.4f\n', a, h);
9 plotbody(U, sig, x0, [-4 4], [-3 3]);
10 title(sprintf('U_\infty=1, \sigma=1, x_0=1'));
11
12 % (c) Submarine: L = 50 m, t = 10 m, U = 5 m/s -> a = 25, h = 5
13 U = 5; a = 25; h = 5;
14
15 % The stagnation condition x0^2 + sig*x0/(pi*U) = a^2 gives sigma directly,
16 % so only x0 is left to find and a single fzero is enough.
17 sigma_of = @(x) pi*U*(a^2 - x.^2)./x;
18 resid = @(x) U*h + sigma_of(x)/pi.*atan(h./x) - sigma_of(x)/2;
19
20 x0 = fzero(resid, [1 0.999*a]);
21 sig = sigma_of(x0);
22 fprintf('(c) sigma = %.2f m^2/s, x0 = %.2f m\n', sig, x0);
23
24 plotbody(U, sig, x0, [-32 32], [-10 10]);
25 title(sprintf('\sigma=%.2f, x_0=%.2f, U_\infty=5', sig, x0));
26
27 %%
28 function plotbody(U, sig, x0, xl, yl)
29     [X, Y] = meshgrid(linspace(xl(1),xl(2),400), linspace(yl(1),yl(2),400));
30     psi = U*Y + sig/(2*pi)*(atan2(Y,X+x0) - atan2(Y,X-x0));
31     a = sqrt(x0^2 + sig*x0/(pi*U));
32
33     figure; hold on;
34     contour(X, Y, psi, 40, 'b');
35     contour(X, Y, psi, [0 0], 'r', 'LineWidth', 2); % body + dividing streamline
36     plot([-x0 x0], [0 0], 'ko', 'MarkerFaceColor', 'k', 'MarkerSize', 5);
37     plot([-a a], [0 0], 'go', 'MarkerFaceColor', 'g', 'MarkerSize', 6);
38     axis equal; box on; xlabel('x'); ylabel('y');
39     legend('streamlines', 'body / stagnation streamline', 'source & sink', ...
40           'stagnation points', 'Location', 'northeast');
41 end
```

A.2 Task 2

```
1 %% Task 2 - Conformal mapping to a Joukowski aerofoil (SID: N7=3, N8=7, N9=6)
2 clear; clc; close all;
3
4 z0 = -0.10 + 0.06i; % x0 = -(N7+N8)/100, y0 = N9/100
5 U = 1; al = deg2rad(15);
6
7 % (a) the circle must pass through the critical point z2 = 1
8 R = abs(1 - z0);
9 bt = -angle(1 - z0); % camber offset angle
10 fprintf('(a) R = %.4f, beta = %.3f deg\n', R, rad2deg(bt));
11
12 % (b) arbitrary circulation
13 Gb = pi*U*R;
14 plotz2(Gb, U, R, al, z0);
15 title(sprintf('\Gamma = \pi U R = %.3f (arbitrary)', Gb));
```

```

16
17 % (c) Kutta condition
18 G = 4*pi*U*R*sin(al + bt);
19 fprintf('(c) Gamma = 4*pi*U*R*sin(alpha+beta) = %.4f\n', G);
20 plotz2(G, U, R, al, z0);
21 title(sprintf('Kutta \Gamma = %.3f', G));
22
23 %% aerofoil and streamlines in the z4-plane
24 th = linspace(0, 2*pi, 401); th(end) = []; % drop the duplicate endpoint (0 = 2*pi)
25 r = [linspace(R, 1.3*R, 60), logspace(log10(1.3*R), log10(12*R), 300)];
26 [TH, RR] = meshgrid(th, r);
27 z1 = RR.*exp(1i*TH);
28 psi = phi(z1, U, R, G, al);
29
30 z2 = z1 + z0;
31 z4 = (z2 + 1./z2)*exp(-1i*al);
32 b = R*exp(1i*th) + z0;
33 b = (b + 1./b)*exp(-1i*al); % aerofoil outline
34 psiB = phi(R, U, R, G, al); % psi on the body surface
35
36 F = scatteredInterpolant(real(z4(:)), imag(z4(:)), psi(:), 'linear', 'none');
37 [X, Y] = meshgrid(linspace(-3,3,400), linspace(-2,2,300));
38 V = F(X, Y);
39
40 figure; hold on;
41 contour(X, Y, V, levels(V, 50), 'b'); % streamlines
42 contour(X, Y, V, [psiB psiB], 'r', 'LineWidth', 1.8); % stagnation streamline
43 fill(real(b), imag(b), 'k');
44 axis equal; box on; xlim([-3 3]); ylim([-2 2]);
45 xlabel('Re(z_4)'); ylabel('Im(z_4)');
46 title(sprintf('Aerofoil in z_4-plane, \alpha=15^\circ'));
47
48 %% (d) surface pressure
49 th = linspace(-pi, pi, 6000);
50 th(abs(angle(exp(1i*(th + bt)))) < deg2rad(0.15)) = []; % skip the TE singular point
51 z1 = R*exp(1i*th); z = z1*exp(-1i*al); z2 = z1 + z0;
52 Vj = (U*(1 - R^2./z.^2) + 1i*G./(2*pi*z)) .* (z2.^2./(z2.^2 - 1));
53 Cp = 1 - abs(Vj).^2;
54
55 % Cp at the trailing edge is a removable 0/0 singularity - take the two-sided limit
56 eps_ = 1e-6;
57 Vte = 0;
58 for d = [-eps_ eps_]
59     z1t = R*exp(1i*(-bt+d)); zt = z1t*exp(-1i*al); z2t = z1t + z0;
60     Vte = Vte + 0.5*(U*(1 - R^2/z2t^2) + 1i*G/(2*pi*z2t))*(z2t^2/(z2t^2 - 1));
61 end
62 CpTE = 1 - abs(Vte)^2;
63 fprintf('(d) Cp at the trailing edge = %.4f\n', CpTE);
64
65 % chordwise coordinate: project the surface onto the LE-TE line
66 z4 = (z2 + 1./z2)*exp(-1i*al);
67 zTE = 2*exp(-1i*al);
68 zLE = (2*z0 - 1) + 1/(2*z0 - 1); zLE = zLE*exp(-1i*al);
69 cc = abs(zTE - zLE); u = (zTE - zLE)/cc;
70 s = real((z4 - zLE)*conj(u))/cc;
71
72 % Split into upper/lower by which side of the trailing-edge angle theta sits on
73 % (NOT by the sign of the chord-normal offset: close to the sharp trailing edge
74 % the surface dips slightly across the chord line, which would misclassify a
75 % few points there). Sort each side by the *unwrapped* angle from the trailing
76 % edge so the two halves of the -180/+180 deg branch cut are stitched back
77 % together correctly.
78 rel = angle(exp(1i*(th - (-bt))));
79 up = rel > 0;
80 [~, iu] = sort(rel(up)); su = s(up); su = su(iu); Cu = Cp(up); Cu = Cu(iu);

```

```

81 [~, il] = sort(rel(~up)); sl = s(~up); sl = sl(il); Cl = Cp(~up); Cl = Cl(il);
82
83 figure;
84 for k = 1:2
85     subplot(1,2,k); hold on;
86     plot(su, Cu, 'b', 'LineWidth', 1.4);
87     plot(sl, Cl, 'r', 'LineWidth', 1.4);
88     plot([0 1], [CpTE CpTE], 'k:');
89     set(gca, 'YDir', 'reverse'); grid on; box on;
90     xlabel('x/c'); ylabel('C_p');
91     if k == 1
92         title('Full range (LE suction peak)');
93         legend('Upper', 'Lower', 'C_p(TE)', 'Location', 'southeast');
94     else
95         ylim([-3 3]); title('Zoomed view');
96     end
97 end
98 sgtitle('C_p on the aerofoil surface, \alpha=15^\circ');
99
100 %% Functions
101 function p = phi(z1, U, R, G, al)
102 % Stream function of the flow past the cylinder, written in z1 coordinates.
103     z = z1*exp(-1i*al);
104     p = imag(U*(z + R^2./z) + 1i*G/(2*pi)*log(z));
105 end
106
107 function L = levels(V, n)
108 % n contour levels spanning the 1st-99th percentile of the visible data,
109 % so the streamlines stay evenly spaced near the body instead of bunching up.
110     v = sort(V(~isnan(V)));
111     lo = v(max(1, round(0.01*numel(v))));
112     hi = v(min(numel(v), round(0.99*numel(v))));
113     L = linspace(lo, hi, n);
114 end
115
116 function plotz2(G, U, R, al, z0)
117     [X, Y] = meshgrid(linspace(-3,3,400), linspace(-2.2,2.2,400));
118     z1 = X + 1i*Y - z0;
119     psi = phi(z1, U, R, G, al);
120     psi(abs(z1) < R) = NaN;
121
122     figure; hold on;
123     contour(X, Y, psi, 45, 'b');
124     psiB = phi(R, U, R, G, al);
125     contour(X, Y, psi, [psiB psiB], 'r', 'LineWidth', 1.8);
126     th = linspace(0, 2*pi, 200);
127     fill(real(z0)+R*cos(th), imag(z0)+R*sin(th), 'k');
128     axis equal; box on; xlabel('Re(z_2)'); ylabel('Im(z_2)');
129 end

```

A.3 Task 3

```

1 %% Task 3 - Thin aerofoil theory (SID: N7=3, N8=7, N9=6)
2 clear; clc; close all;
3
4 c = 1; al = deg2rad(3); % alpha = N7 degrees
5 yc = @(x) 0.195*(c-x).^2.*x/c^3; % 3*(N8+N9)/200 = 0.195
6 dydx = @(x) 0.195*(c-x).*(c-3*x)/c^3;
7
8 % (a) camber line
9 x = linspace(0, c, 200);
10 figure; plot(x, yc(x), 'b', 'LineWidth', 2);
11 axis equal; grid on; box on;

```

```

12 xlabel('x/c'); ylabel('y/c'); title('Camber line, c = 1');
13
14 % (b)-(d) Glauert coefficients, x = c/2*(1 - cos(theta))
15 g = @(t) dydx(c/2*(1 - cos(t)));
16 I0 = integral(g, 0, pi);
17 I1 = integral(@(t) g(t).*cos(t), 0, pi);
18 I2 = integral(@(t) g(t).*cos(2*t), 0, pi);
19
20 aL0 = (I0 - I1)/pi;
21 A0 = a1 - I0/pi; A1 = 2*I1/pi; A2 = 2*I2/pi;
22 c1 = pi*(2*A0 + A1);
23 cm = -pi/4*(A1 - A2);
24
25 fprintf('(b) alpha_L0 = %.4f deg\n', rad2deg(aL0));
26 fprintf('(c) c1 = %.4f\n', c1);
27 fprintf('(d) cm_c/4 = %.4f\n', cm);
28
29 % numerical check - lumped vortex model with 10 panels
30 [cl10, cm10] = lvm(yc, al, 10, c);
31 a0 = fzero(@(a) lvm(yc, a, 10, c), [-deg2rad(5) deg2rad(5)]);
32 fprintf('\nLVM (10 panels): c1 = %.4f, cm_c/4 = %.4f, alpha_L0 = %.4f deg\n', ...
33         cl10, cm10, rad2deg(a0));
34
35 %%%
36 function [c1, cm4] = lvm(yc, al, N, c)
37     xb = linspace(0, c, N+1); zb = yc(xb);
38     dx = diff(xb); dz = diff(zb); th = atan2(dz, dx);
39     xv = xb(1:N) + 0.25*dx; zv = zb(1:N) + 0.25*dz; % vortex at 1/4 panel
40     xc = xb(1:N) + 0.75*dx; zc = zb(1:N) + 0.75*dz; % control pt at 3/4 panel
41
42     A = zeros(N);
43     for i = 1:N
44         dxv = xc(i) - xv; dzv = zc(i) - zv; r2 = dxv.^2 + dzv.^2;
45         A(i,:) = (-dzv./r2)*(-sin(th(i))) + (dxv./r2)*cos(th(i));
46     end
47     A = A/(2*pi);
48     G = A\((sin(th - al))'); % U = 1
49
50     c1 = -2*sum(G)/c; % CCW-positive circulation -> L' = -rho*U*G
51     cmLE = 2*sum(xv(:).*G)/c^2;
52     cm4 = cmLE + c1/4;
53 end

```

A.4 Task 4

```

1 %%% Task 4 - 2D constant-strength panel method
2 clear; clc; close all;
3
4 %%% (a)(b) cylinder, 64 source panels
5 th = linspace(0, 2*pi, 65)'; % CCW ordering -> outward normals
6 xn = cos(th); yn = sin(th);
7 [xc, yc, S, ph, nx, ny] = geom(xn, yn);
8 [I, J] = infl(xc, yc, xn, yn, S, ph, 'source');
9
10 sig = I\(-nx); % U = 1, alpha = 0
11 Cp = 1 - (cos(ph) + J*sig).^2;
12 tc = atan2(yc, xc);
13 fprintf('(a) max |Cp - theory| = %.2e\n', max(abs(Cp - (1 - 4*sin(tc).^2))));
14
15 [X, Y] = meshgrid(linspace(-5,5,300));
16 [u, v] = vel(X, Y, xn, yn, S, ph, sig, 0);
17 u = u + 1;
18 in = X.^2 + Y.^2 < 1; u(in) = NaN; v(in) = NaN;

```

```

19 figure('Position', [100 100 1100 480]);
20 subplot(1,2,1); hold on; box on;
21 h = streamslice(X, Y, u, v, 1.5); set(h, 'Color', 'b');
22 fill(xn, yn, 'k');
23 axis equal; xlim([-5 5]); ylim([-5 5]);
24 xlabel('x'); ylabel('y'); title('Streamlines');
25
26
27 subplot(1,2,2); hold on; box on;
28 plot(rad2deg(tc), Cp, 'bo', 'MarkerSize', 4, 'MarkerFaceColor', 'b');
29 t = linspace(-180, 180, 300);
30 plot(t, 1 - 4*sind(t).^2, 'r', 'LineWidth', 1.5);
31 grid on; xlabel('\theta (deg)'); ylabel('C_p');
32 legend('Panel method', 'Theory C_p = 1 - 4sin^2\theta', 'Location', 'south');
33 title('C_p comparison');
34 sgtitle('Source panel method, cylinder, N = 64');
35
36 %% (c) NACA0021 at alpha = 0, source panels only
37 [xn, yn] = naca(0.21, 100);
38 [xc, yc, S, ph, nx, ny] = geom(xn, yn);
39 [I, J] = infl(xc, yc, xn, yn, S, ph, 'source');
40 sig = I\(-nx);
41 Cp0 = 1 - (cos(ph) + J*sig).^2;
42
43 figure('Position', [100 100 760 520]);
44 plotcp(xc, yc, Cp0, 'xfoil_naca0021_a0.dat', false);
45 title('NACA0021, \alpha=0^\circ, source panels vs XFoil');
46
47 %% (d)(e)(f) NACA0021 at alpha = 10, source + vortex with Kutta condition
48 al = deg2rad(10); N = numel(xc);
49 [K, L] = infl(xc, yc, xn, yn, S, ph, 'vortex');
50
51 A = [I, sum(K,2); J(1,:) + J(N,:), sum(L(1,:)) + sum(L(N,:))];
52 b = [-cos(al)*nx - sin(al)*ny;
53      -cos(al)*(cos(ph(1)) + cos(ph(N))) - sin(al)*(sin(ph(1)) + sin(ph(N)))];
54 q = A\b; sig = q(1:N); gam = q(N+1);
55
56 Cp = 1 - (cos(al)*cos(ph) + sin(al)*sin(ph) + J*sig + gam*sum(L,2)).^2;
57 cx = sum(-Cp.*nx.*S); cy = sum(-Cp.*ny.*S);
58 cl = cy*cos(al) - cx*sin(al);
59 cd = cx*cos(al) + cy*sin(al);
60 fprintf('\n(f) cl = %.4f (pressure), %.4f (Kutta-Joukowski)\n', cl, -2*gam*sum(S));
61 fprintf(' cd = %.5f (~0, d\'Alembert)\n', cd);
62
63 figure('Position', [100 100 760 520]);
64 plotcp(xc, yc, Cp, 'xfoil_naca0021_a10.dat', true);
65 title(sprintf('NACA0021, \alpha=10^\circ, c_l=%.3f, vs XFoil', cl));
66
67 %% streamlines plus one released from the trailing edge
68 [X, Y] = meshgrid(linspace(-1.5,2.5,350), linspace(-1.5,1.5,350));
69 [u, v] = vel(X, Y, xn, yn, S, ph, sig, gam);
70 u = u + cos(al); v = v + sin(al);
71 in = inpolygon(X, Y, xn, yn); u(in) = NaN; v(in) = NaN;
72
73 figure('Position', [100 100 860 580]); hold on; box on;
74 h = streamslice(X, Y, u, v, 1.3); set(h, 'Color', 'b');
75
76 p = [1 + 1e-4*cos(al), 1e-4*sin(al)]; P = p;
77 for k = 1:250 % RK4 along the streamline
78     f1 = tev(p, xn,yn,S,ph,sig,gam,al);
79     f2 = tev(p+0.01*f1, xn,yn,S,ph,sig,gam,al);
80     f3 = tev(p+0.01*f2, xn,yn,S,ph,sig,gam,al);
81     f4 = tev(p+0.02*f3, xn,yn,S,ph,sig,gam,al);
82     p = p + 0.02*(f1 + 2*f2 + 2*f3 + f4)/6; P(end+1,:) = p;
83     if p(1) > 2.4, break; end

```

```

84 end
85 plot(P(:,1), P(:,2), 'r', 'LineWidth', 2);
86 fill(xn, yn, 'k');
87 axis equal; xlim([-1.5 2.5]); ylim([-1.5 1.5]);
88 xlabel('x/c'); ylabel('y/c');
89 legend('', 'streamline from the trailing edge', 'Location', 'northwest');
90 title(sprintf('Streamlines, \\alpha=10^\\circ, c_l=%.3f', c_l));
91
92 %% Functions
93 function [xc, yc, S, ph, nx, ny] = geom(xn, yn)
94     xc = (xn(1:end-1) + xn(2:end))/2; yc = (yn(1:end-1) + yn(2:end))/2;
95     dx = diff(xn); dy = diff(yn);
96     S = hypot(dx, dy); ph = atan2(dy, dx);
97     nx = sin(ph); ny = -cos(ph); % outward normal for CCW node ordering
98 end
99
100 function [An, At] = infl(xc, yc, xn, yn, S, ph, kind)
101 % Normal (An) and tangential (At) influence of unit-strength panels.
102 N = numel(xc); An = zeros(N); At = zeros(N);
103 isSource = strcmp(kind, 'source');
104 for i = 1:N
105     for j = 1:N
106         if i == j % self-induced terms
107             if isSource, An(i,j) = pi; else, At(i,j) = pi; end
108             continue
109         end
110         cp = cos(ph(j)); sp = sin(ph(j));
111         a = (xc(i)-xn(j))*cp + (yc(i)-yn(j))*sp; % panel-local coords
112         b = -(xc(i)-xn(j))*sp + (yc(i)-yn(j))*cp;
113         lg = 0.5*log((a^2 + b^2)/((a-S(j))^2 + b^2));
114         dt = atan2(b, a-S(j)) - atan2(b, a);
115         if isSource
116             p = lg; q = dt;
117         else
118             p = -dt; q = lg;
119         end
120         u = p*cp - q*sp; v = p*sp + q*cp; % back to global axes
121         An(i,j) = u*sin(ph(i)) - v*cos(ph(i));
122         At(i,j) = u*cos(ph(i)) + v*sin(ph(i));
123     end
124 end
125 An = An/(2*pi); At = At/(2*pi);
126 end
127
128 function [u, v] = vel(X, Y, xn, yn, S, ph, sig, gam)
129 % Velocity induced at arbitrary field points by all panels.
130 u = zeros(size(X)); v = zeros(size(X));
131 for j = 1:numel(S)
132     cp = cos(ph(j)); sp = sin(ph(j));
133     a = (X-xn(j))*cp + (Y-yn(j))*sp;
134     b = -(X-xn(j))*sp + (Y-yn(j))*cp;
135     lg = 0.5*log((a.^2 + b.^2)/((a-S(j)).^2 + b.^2));
136     dt = atan2(b, a-S(j)) - atan2(b, a);
137     p = sig(j)*lg - gam*dt; % source part + vortex part
138     q = sig(j)*dt + gam*lg;
139     u = u + (p*cp - q*sp)/(2*pi);
140     v = v + (p*sp + q*cp)/(2*pi);
141 end
142 end
143
144 function f = tev(p, xn, yn, S, ph, sig, gam, al)
145     [u, v] = vel(p(1), p(2), xn, yn, S, ph, sig, gam);
146     f = [u + cos(al), v + sin(al)];
147 end
148

```

```

149 function [xn, yn] = naca(t, n)
150 % NACA 00tt, cosine spaced, CCW from the trailing edge, closed TE.
151 x = (1 - cos(linspace(0, pi, n+1')))/2;
152 yt = 5*t*(0.2969*sqrt(x) - 0.1260*x - 0.3516*x.^2 + 0.2843*x.^3 - 0.1036*x.^4);
153 xn = [flipud(x); x(2:end)];
154 yn = [flipud(yt); -yt(2:end)];
155 end
156
157 function plotcp(xc, yc, Cp, file, splitColour)
158 % splitColour = false -> both surfaces blue (they coincide at alpha = 0)
159 % splitColour = true -> blue upper, green lower
160 up = yc >= 0;
161 [~, a] = sort(xc(up)); xu = xc(up); cu = Cp(up);
162 [~, b] = sort(xc(~up)); xl = xc(~up); cd_ = Cp(~up);
163
164 hold on; box on;
165 if splitColour
166     plot(xu(a), cu(a), 'b', 'LineWidth', 1.5);
167     plot(xl(b), cd_(b), 'g', 'LineWidth', 1.5);
168     names = {'Panel method (upper)', 'Panel method (lower)'};
169 else
170     plot(xu(a), cu(a), 'b', 'LineWidth', 1.6);
171     plot(xl(b), cd_(b), 'b', 'LineWidth', 1.6);
172     names = {'Panel method', ''};
173 end
174
175 if isfile(file)
176     D = readmatrix(file, 'NumHeaderLines', 1, 'FileType', 'text');
177     plot(D(:,1), D(:,2), 'r--', 'LineWidth', 1.3);
178     names{end+1} = 'XFOIL (inviscid)';
179 else
180     warning('%s not found - plotting the panel method only.', file);
181 end
182
183 set(gca, 'YDir', 'reverse'); grid on;
184 xlabel('x/c'); ylabel('C_p');
185 legend(names{:}, 'Location', 'best');
186 end

```

Statement of AI Use

AI (Claude) was used to help with code structure and debugging assistance and to improve the clarity and structure of my written explanations in this report.